To test your understanding of the Markov property, consider each control problem or decision task in the following list and see if it has the Markov property or not:

- Driving a car
- Deciding whether to invest in a stock or not
- Choosing a medical treatment for a patient
- Diagnosing a patient's illness
- Predicting which team will win in a football game
- Choosing the shortest route (by distance) to some destination
- Aiming a gun to shoot a distant target

Okay, let's see how you did. Here are our answers and brief explanations:

- Driving a car can generally be considered to have the Markov property because you don't need to know what happened 10 minutes ago to be able to optimally drive your car. You just need to know where everything is right now and where you want to go.
- Deciding whether to invest in a stock or not does not meet the criteria of the Markov property since you would want to know the past performance of the stock in order to make a decision.
- Choosing a medical treatment seems to have the Markov property because you don't need to know the biography of a person to choose a good treatment for what ails them right now.
- In contrast, *diagnosing* (rather than treating) would definitely require knowledge of past states. It is often very important to know the historical course of a patient's symptoms in order to make a diagnosis.
- Predicting which football team will win does not have the Markov property, since, like the stock example, you need to know the past performance of the football teams to make a good prediction.
- Choosing the shortest route to a destination has the Markov property because you just need to know the distance to the destination for various routes, which doesn't depend on what happened yesterday.
- Aiming a gun to shoot a distant target also has the Markov property since all you need to know is where the target is, and perhaps current conditions like wind velocity and the particulars of your gun. You don't need to know the wind velocity of yesterday.